Lakshman Chelliah

Software Engineering Developer · Embedded systems, web products & AI

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PROFILE

Software engineering graduate with production experience across vehicle electronics, photonic networking, secure government platforms, and test automation. Builds embedded and real-time systems alongside web products and multiplayer games. Uses research, practical tooling, and LLM-assisted workflows to learn unfamiliar technologies, implement solutions, and verify results.

EXPERIENCE

Software Engineering Developer
May 2026 – Present

General Dynamics Mission Systems–Canada · Vetronics · Smart Displays

- Software development within the Smart Displays team for vehicle electronics.

Intercept System Developer
May 2025 – Aug 2025

Royal Canadian Mounted Police · Technical Operations

- Contributed automated data-purging features for sensitive information in accordance with stakeholder requirements.
- Investigated intermittent CI/CD failures and environment-specific issues; tested and validated application changes.

Photonics Software Developer
Sept 2024 – Dec 2024

Ciena

- Contributed to FPGA upgrade workflows for amplifiers and ROADMs on the 6500 Reconfigurable Line System.
- Developed PRAM-based diagnostic logging and validated recovery from fibre cuts and loss-of-signal events in hardware-in-the-loop environments.

Web Developer
May 2024 – Aug 2024

Royal Canadian Mounted Police · Technical Operations

- Upgraded an Angular application from v13 to v18, refactoring legacy components and resolving compatibility issues.
- Developed unit and web UI tests to check feature parity; contributed code reviews and changes through deployment.

Machine Learning Developer
Jan 2024 – Apr 2024

BlackBerry

- Built a Salesforce data-ingestion pipeline to produce structured datasets for predictive modelling.
- Evaluated lightweight LLMs for system-log summaries and developed Python/PowerShell automation for Microsoft 365 and Azure licence audits.

Embedded Software Developer
May 2023 – Aug 2023

Ciena

- Resolved Coverity findings involving memory leaks, uninitialized variables, and buffer overflows.
- Built Python tooling to compare software-load boot times across RLS modules and recreate long-running customer defects.
- Automated log analysis and engineering notifications during defect reproduction.

Automation Software Engineer
Jan 2022 – Sept 2022

Telepin Software Systems

- Built automated regression tests for payment, refund, and commission workflows.
- Translated business rules and edge cases into API and UI validation using Java, JMeter, and SQL.

Lakshman Chelliah

Selected projects, education & technical skills

SELECTED PROJECTS

ESSTA

HTML / CSS · JavaScript · D3

Built a responsive community website with heritage visualization, configurable content, and a membership application workflow.

Conqr

Next.js · PartyKit · TypeScript

Built a multiplayer strategy game with authoritative PartyKit rooms, a pure rules engine, seeded randomness, reconnection, and private player state.

BeyWeb

Three.js · Cannon-es · Node.js

Built a 3D browser arena game with shared physics simulation, authoritative WebSockets, client prediction/reconciliation, and persistent customization/progression.

QNX RC Vehicle

Jan – Apr 2026

University team project integrating QNX IPC, drive-source arbitration, sensor processes, emergency stop, and a command watchdog on a Raspberry Pi RC vehicle.

FileTriage

Electron · React · TypeScript

Built an Electron/React media-review tool with previews, exact duplicate identification, Recycle Bin integration, and undo.

IoT Trust Management · Capstone team project

Oct 2025 – Apr 2026

Neural-network trust scoring for ON–OFF attack detection, behavioral features, SHAP explanations, repeatable data snapshots, and an LLM-based security-summary layer.

Automotive Firmware Reverse Engineering

Mar – Apr 2026

PowerPC ECU analysis with Ghidra and WinOLS; investigated LIN and NVRAM routines and implemented a Tow/Haul persistence patch with assembly-level hooks and scripted analysis.

EDUCATION

Carleton University

Bachelor of Engineering · Software Engineering with Co-op

Sept 2021 – April 2026

TECHNICAL SKILLS

Systems & embedded: C, C++, QNX, VxWorks, Embedded Linux, ARM, IPC, UART / SPI / I2C

Web & products: TypeScript, JavaScript, React / Next.js, Angular, Node.js, Spring Boot, WebSockets, SQL

AI & automation: Python, LLM-assisted workflows, Pandas / NumPy, SHAP, Ghidra automation, PowerShell

Verification & delivery: Git, GitLab CI/CD, Docker, Playwright, Vitest, Coverity, HIL testing, Wireshark